Project Development Phase

Phase 5: Implementation & Development

Project Name	Automated Network Request Management in ServiceNow
Phase	Phase 5: Project Development
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Platform / Domain	ServiceNow / IT Service Management (ITSM)
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1. Introduction

The Automated Network Request Management System was successfully developed on a ServiceNow personal developer instance. The primary objective of this implementation phase was to engineer a robust, automated lifecycle for infrastructure requests — including VPN provisioning, firewall modifications, and network access changes — by completely eliminating error-prone email loops and manual hand-offs. The solution delivers cross-functional tracking via the consumer-grade Service Portal, secure Access Control Lists (ACLs), and automated operational orchestration powered by Flow Designer.

2. Milestone 1 — Application Setup & Portal Initialization

Objective

Establish the development environment framework and front-end service catalog structure.

Activities Performed

- Initialized the system architecture to support standard network operations catalog items natively.
- Designed the front-end catalog taxonomy under the Service Catalog → Network Requests menu hierarchy.
- Configured user portal accessibility criteria to display catalog items cleanly via standard Service Portal widgets.
- Established the 8 network request catalog item placeholders and assigned them to the Network Requests category.

3. Milestone 2 — Database & Schema Configuration

Objective

Create and index a custom structural logging repository to track historical data transactions alongside baseline ITSM tables.

Activities Performed

- Constructed the custom Network Database storage table: u_network_database.
- Provisioned critical tracking columns: u_request_number (String), u_requested_for (String), u_device_type (String), u_work_status (String), u_updates (Integer).
- Set default display mappings and field lengths to preserve structural normalization.
- Validated table dictionary structure and confirmed field indexing for query performance.

4. Milestone 3 — User, Role-Based Access Control & Security Matrix

Objective

Define strict security governance profiles via Access Control Lists (ACLs) to regulate read, write, create, and delete operations on all data tables.

Activities Performed

- Provisioned test user records within the sys_user identity engine and assigned them to appropriate role groups.
- Generated and configured explicit ACL rules across all transactional vectors: create, read, write, delete.
- Structured access constraints ensuring requesters have visibility only into their own submitted items.
- Granted sn_network_engineer role full write access over assigned task records while restricting access to unrelated group queues.
- Validated all four ACL constraint rules against the u_network_database and sc_req_item tables.

5. Milestone 4 — User Interface & Dynamic Form Engineering

Objective

Design dynamic Service Portal request forms using advanced field-level controls and interactive validation logic.

Activities Performed

- Engineered individual functional catalog forms for all 8 network request types.
- Implemented dynamic mandatory field indicators to block incomplete form submissions.
- Configured field layouts to capture: requester identity, email ID, phone number, affected device type, technical parameters (IP addresses, VLAN IDs, port numbers), business justification, and proof-of-authorization document attachment.
- Applied regex-based Client Scripts to validate IP address format (e.g., 192.168.x.x), CIDR notation, and port number ranges before form submission.
- Implemented UI Policies to conditionally show or hide fields based on user selections (e.g., VPN type toggle reveals profile-specific fields).

6. Milestone 5 — Workflow Orchestration & Notification Engine

Objective

Configure end-to-end Flow Designer workflows linking catalog item submission to automated approval routing, fulfillment task generation, SLA management, and event-driven notifications.

Activities Performed

- Built Flow Designer workflows for all 8 catalog items, with conditional branching for risk-based approval routing.
- Configured Stage 1: Line Manager Approval — request routed to requester's manager; approval email dispatched; SLA-adjacent tracking initiated.
- Configured Stage 2 (high-risk items): Network Security Review — activated automatically upon manager approval for firewall and VLAN items.
- Configured Stage 3: Fulfillment Task Generation — sc_task auto-created and routed to Network Operations Group queue upon final approval sign-off.
- Configured SLA Management — SLA clock activates upon sc_task creation with escalation triggers at 50%, 75%, and 100% of the breach window.
- Configured Notification Engine — automated email templates dispatched at: submission, pending approval, approval granted, approval rejected, task assigned, and request closed.
- Validated end-to-end workflow execution through full lifecycle test scenarios for each catalog item type.

7. Development Summary

Milestone	Description	Status	Completion Date

Milestone 1	Application Setup & Portal Initialization	Complete	22–23 June 2026
Milestone 2	Database & Schema Configuration	Complete	23–24 June 2026
Milestone 3	RBAC & Security Matrix	Complete	24–25 June 2026
Milestone 4	UI & Dynamic Form Engineering	Complete	25–26 June 2026
Milestone 5	Workflow Orchestration & Notifications	Complete	26–27 June 2026